

Jenish Vavadiya

vavadiyajenish25@gmail.com | +91 8160159516 | LinkedIn | GitHub | LeetCode

PROFESSIONAL SUMMARY

Quantitative Researcher and Blockchain Engineer with hands-on experience in DeFi markets, MEV research, and algorithmic trading systems. Skilled in analyzing market behavior, liquidity flows, pricing inefficiencies, and execution mechanisms across on-chain environments. Currently building Solana-based arbitrage infrastructure involving real-time data analysis, route optimization, and market monitoring. Interested in prediction markets, probabilistic forecasting, market design, and information aggregation, with a strong foundation in Rust, financial systems, and data-driven decision making.

TECHNICAL SKILLS

Languages:	Rust, JavaScript, TypeScript, Python (NumPy, pandas, scikit-learn), Java, SQL, Solidity
Quantitative & Probabilistic:	Bayesian Reasoning, Probabilistic Forecasting, Calibration Evaluation, Time-Series Analysis, Backtesting, Simulation, Statistical Modeling
Market Mechanisms & DeFi:	AMM / CLMM Design, On-chain Order Books, Liquidity Pool Mechanics, Incentive & Fee Structures, MEV Strategies, Prediction Market Design, Token-2022, SPL Tokens
MEV & Performance:	Jito Bundles, Flashbots, MEV-Boost, SUAVE, Rollup-Boost, Sandwich / Liquidation Strategies, CU Optimization, Priority Fees, Transaction Simulation, Sealevel Parallelism, Durable Nonces
Data & Backend:	Data Pipelines, Dune Analytics, Node.js, Express, Spring Boot, REST APIs, MongoDB, MySQL, React, Recoil, Tailwind CSS
Tools:	GitHub, VS Code, Cursor AI, IntelliJ IDEA, Postman, Solpg, Geyser Yellowstone gRPC

EXPERIENCE

Prediction Markets & DeFi Quant Intern Whistler Trading | Nov 2025 – Present

- Designed and analyzed AMM / CLMM pool mechanics across Raydium, Orca, Meteora, and PumpFun — modeled liquidity depth, price-impact curves, and slippage to identify mispricing signals, directly analogous to probability aggregation in prediction markets.
- Built a probabilistic profit-estimation pipeline that simulates each swap leg off-chain, applies Bayesian slippage models, deducts fee/tip costs, and filters only bundles above a configurable net-profit floor — demonstrating calibrated forecasting under uncertainty.
- Engineered a priority-fee oracle reading on-chain recent fee percentiles (p75) and auto-adjusting bids based on mempool congestion and margin thresholds — an incentive mechanism grounded in game-theoretic bidding strategy.
- Implemented dynamic Compute Unit budget selection by profiling per-route transaction requirements, reducing fee overpay while maximising validator inclusion rate.
- Integrated Jito block-engine bundles for atomic multi-leg execution: forward swap, backward swap, and Jito tip — eliminating sandwich exposure on profitable routes.
- Stack: Rust, Solana, Jito SDK, Geyser Yellowstone gRPC, Pinocchio.

MEV Research & Market Mechanism Intern Niqox Tech LLP, Surat | Jun 2025 – Nov 2025

- Researched and implemented MEV strategies — sandwich attacks, liquidation monitoring, front-/back-running — applying game-theoretic reasoning to adversarial multi-agent market settings.
- Studied Flashbots, MEV-Boost, SUAVE, and Rollup-Boost ecosystems for cross-chain MEV protection, developing intuition for incentive-aligned mechanism design across heterogeneous trading venues.
- Analysed Sealevel parallelism, durable nonces, and validator leader scheduling to optimise transaction landing rates; quantified throughput trade-offs using real on-chain data.

Blockchain & Smart Contract Developer Intern Zerobit Infotech, Surat | Jun 2024 – Dec 2024

- Developed and deployed smart contracts on Ethereum and Solana; designed AMM liquidity-pool mechanics and integrated on-chain price feeds, building foundational understanding of DeFi protocol incentive structures.
- Built NFT marketplace with minting, storage, and deployment pipelines using Web3.js and Anchor — experience directly transferable to on-chain prediction-market contract development.

- Demonstrated expertise in Solidity, Rust, and DeFi protocols including AMM liquidity pool design and on-chain price feed integration.

Java Backend Developer Intern

ADD Zero Group, Surat | May 2023 – Apr 2024

- Built functional REST APIs using Java, Spring Boot, and SQL for real-world production projects; strong proficiency in Data Structures and Algorithms; active LeetCode practitioner.

PROJECTS

Solana Arbitrage Bot — Multi-Venue Price Aggregation (Self-Project)

- Built a production arbitrage bot scanning 3-leg routes across Jupiter-aggregated DEX pools; route-scoring logic is structurally equivalent to aggregating probabilistic signals from multiple prediction-market venues to find mispriced outcomes.
- CU Optimisation: profiles each instruction set, sets exact ComputeBudget limits, and batches Address Lookup Table (ALT) creations to stay within per-transaction CU caps.
- MEV Defence: monitors competing bundles; applies tip-escalation strategy to outbid adversaries while maintaining profitability thresholds — a game-theoretic bidding model under incomplete information.
- Stack: Rust, TypeScript, Jupiter API v6, Jito, Solana web3.js.

MEV Research & Validator Optimisation — Market Microstructure Research

- Conducted deep research into Sealevel parallelism, compute bottlenecks, durable nonces, nonce collisions, and validator leader slots for transaction scheduling optimisation.
- Analysed stake delegation and tip-based subsidies across Solana validators using Dune Analytics; built data pipelines to extract behavioural signals and liquidity dynamics from on-chain data.
- Key finding: optimal CU budget = simulated usage + 15% buffer; priority fee at p75 recent fee percentile maximises landing rate vs. cost.

EDUCATION

B.Tech in Computer Science & Engineering

P. P. Savani University | 2022 – 2026

CGPA: 8.04 / 10.0